Reg. No. | | | | | | | | | |

Question Paper Code : 12802

B.E / B.TECH DEGREE EXAMINATION, MAY / JUNE 2017

Third Semester

Common to B.E – Computer Science Engineering & B.Tech – Information Technology

15MA304 – DISCRETE MATHEMATICS

(Regulations: Mepco – R2015)

Duration: 3 Hours Max. : 100 Marks

Answer ALL Questions

PART A – (10 × 2 = 20 Marks)

1. Using set notation, rewrite the following statements "x is a member of A", "F is superset of G".
2. Give an example for a bijective function.
3. What is the truth value of $\forall x\, P(x)$, where $P(x)$ is the statement "$x^2 < 10$" and the domain consists of the positive integers 1,2,3 and 4?
4. Write a note on the technique "proof by contradiction" used to prove theorems.
5. Draw a graph whose degree sequence is 5,2,2,2,2,1.
6. Define, "A Tree, a Spanning tree" of a graph G.
7. Give an example of a Semi group which is not a Monoid.
8. When will a ring be said to be a commutative ring?
9. Draw the Hasse diagram for $(A, \leq)$ where A = {1,2,3,4}.
10. Describe a Complemented lattice with example.

PART B – (5 × 16 = 80 Marks)

11. a) i. Establish $A \cup (B \cup C) = (A \cup B) \cup C$ analytically. And also prove that $A \cap (B \cap C) = (A \cap B) \cap C$ analytically. (8 Marks)

11. a) ii. Let f and g be the functions from the set of integers to the set of integers defined by $f(x) = 2x + 3$ and $g(x) = 3x + 2$. What is the composition of f and g? What is the composition of g and f? Is the composition of f and f invertible? (8 Marks)

OR

11. b) i. Suppose that the relations R_1 and R_2 on a set A are represented by the matrices $M_{R_1} = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$ and $M_{R_2} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix}$, what are the matrices representing $R_1 \cup R_2$, $R_1 \cap R_2$, $R_1 \circ R_2$ and R^{-1}? (8 Marks)

11. b) ii. List all possible functions from $X = \{1,2\}$ to $Y = \{a, b, c\}$ and classify them as One-to-one, Onto, neither One-to-one nor Onto function. (8 Marks)

12. a) i. Show that $\backsim (p \vee (\backsim p \wedge q))$ and $\sim p \wedge \sim q$ are logically equivalent by developing truth table. (8 Marks)

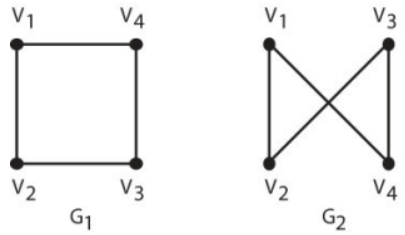
12. a) ii. Show that the hypotheses, "If you send me an e-mail message, then I will finish writing program", "If you do not send me an e-mail message, then I will go to sleep early" and "If I go to sleep early, then I will wake up feeling refreshed" lead to the conclusion "If I do not finish writing the program, I will wake up feeling refreshed" using Rules of Inference for proposition. (8 Marks)

OR

12. b) i. Show that $(p \wedge q) \rightarrow (p \vee q)$ is a tautology (A) without using truth table, (B) using truth table. (8 Marks)

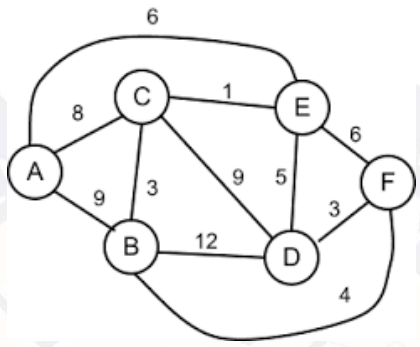
12. b) ii. Show that the premises, "Everyone in this discrete mathematics class has taken a course in computer science" and " Marla is a student in this class" imply the conclusion "Marla has taken a course in computer science" using Rules of Inference for quantifiers. (8 Marks)

13. a) i. Define: Isomorphic graphs. Determine whether the given graphs are isomorphic or not.



(8 Marks)

13. a) ii. Find the minimum spanning tree for the weighted graph shown below by using Kruskal's algorithm.

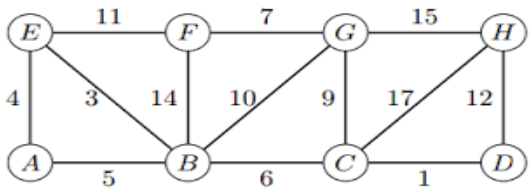


(8 Marks)

OR

13. b) i. Prove that an undirected graph is a tree if and only if there is a unique simple path between any two of its vertices. (8 Marks)

13. b) ii. Use Dijkstra's algorithm to find the shortest path between vertices "A" and "H" in the weighted graph given below:



(8 Marks)

14. a) i. Prove that $M_2(R)$- set of all 2×2 matrices of real numbers is a group with respect to matrix addition. (8 Marks)

14. a) ii. If $(H,*)$ is a subgroup of $(G,*)$, then prove that any two cosets of H in G are either disjoint or identical. (8 Marks)

OR

14. b) i. State and prove the necessary and sufficient condition for a non-empty subset H of a Group $(G,*)$ to be a subgroup of G. (8 Marks)

14. b) ii. If $f:(G,*) \to (G',\Delta)$ is homomorphism with kernel K, prove that the quotient group G/K is isomorphic to homomorphic image of G. (8 Marks)

15. a) i. State and prove modular inequality in a lattice. (8 Marks)

15. a) ii. Verify that whether $\left(D_{35},/\right)$ is a Boolean algebra or not. (8 Marks)

OR

15. b) i. If $(L,\leq)$ is a lattice, prove that $a\wedge a = a, a\vee a = a$ for any $a \in L$. (8 Marks)

15. b) ii. Define Distributive lattice. Prove or disprove that every chain is a distributive lattice. (8 Marks)